



1 (a) State Newton's law of gravitation.

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.....

..... [2]

(b) A planet may be considered as a uniform sphere.

A satellite is in circular orbit of period T around the planet at a height h above the surface. The height of the orbit can be adjusted by use of the satellite's rocket engines.

Fig. 1.1 shows the variation with h of $T^{\frac{2}{3}}$.

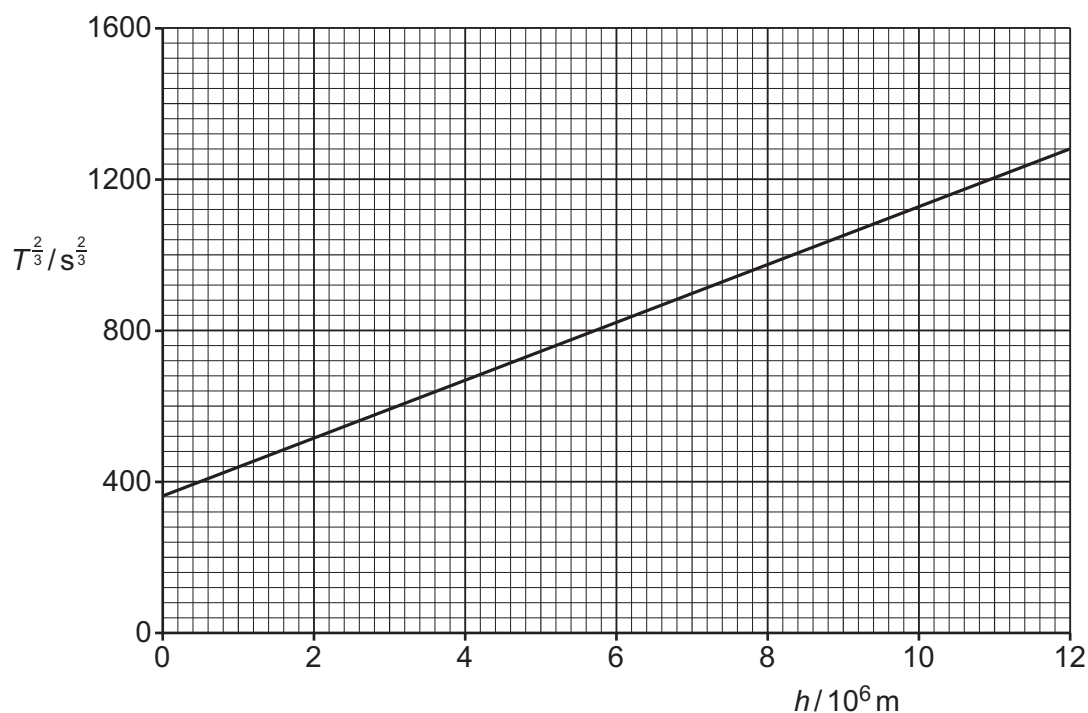


Fig. 1.1

(i) By reference to forces, explain why the orbit of the satellite is circular.

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..... [2]





- (ii) Use Newton's law of gravitation to show that h and T are related by

$$(h + B)^3 = \frac{GA}{4\pi^2} T^2$$

where G is the gravitational constant and A and B are constants that depend on the properties of the planet.

[3]

- (iii) Use the gradient and intercept of the line in Fig. 1.1 to determine values for A and B .
Give units with your answers.

$A = \dots\dots\dots$ unit $\dots\dots\dots$

$B = \dots\dots\dots$ unit $\dots\dots\dots$

[5]

[Total: 12]

